NATIONAL UNIVERSITY BANGLADESH



Syllabus
Bachelor of Science (Honors) Second Year
Department of Biochemistry and Molecular Biology

PROVISIONAL

Effective from the Session: 2024–2025 and onwards

Vision of the University

We expect that the hundredth anniversary of the National University, our commitment to "Progress for Everyone" will be evident in all areas of our operations and stakeholder engagements. We leverage our core strength in education, research, and community engagement to address pressing global issues and foster sustainable development. By 2042, National University aims to strengthen its positions as Bangladesh's top academic institution and rank among the world's leading universities. Our focus will be on expanding expertise, building strategic partnerships, and increasing our global influence. Our services will be measured by the achievements of the individuals and communities we serve.

Mission of the University

To promote "Progress for Everyone" by fostering transformative education, conducting pioneering research, and encouraging meaningful collaboration with individuals, communities, and partners both in Bangladesh and around the world. We want to develop skilled graduates with strong ethical values, scientific integrity, leadership, and lifelong learning abilities. Our mission to create an inclusive and supportive learning environment that inspires creativity, innovation, and professional excellence.

Name of the Program: B.Sc. (Honours) in Biochemistry and Molecular Biology

Vision of the Department

To become a center of excellence in Biochemistry and Molecular Biology through quality education, innovative research, and community engagement, producing competent graduates and scientific leaders who contribute to advancements in healthcare, biotechnology, and sustainable development at national and global levels.

Mission of the Department

The mission of the department is:

1. To provide high-quality education in biochemistry and molecular biology through modern curricula, laboratory-based learning, and critical thinking.
2. To promote innovative and interdisciplinary research that addresses national and global health, environmental, and biotechnological challenges.
3. To foster collaborations with academic institutions, research organizations, health-care sectors, and industries for knowledge exchange and technological advancement.
4. To encourage the application of scientific knowledge for improving public health, disease prevention, environmental sustainability, and socioeconomic development.

Program Educational Objectives (PEOs)

Graduates of the B.Sc. (Honours) Program in Biochemistry and Molecular Biology are expected to achieve the following within four years after graduation:

PEO 1: Demonstrate a strong foundation in biochemistry and molecular biology and apply scientific knowledge and laboratory skills to solve problems in healthcare, biotechnology, pharmaceuticals, agriculture, environmental sciences, and related fields.

PEO 2: Engage in lifelong learning, higher education, research, innovation, and professional development through scientific inquiry, multidisciplinary collaboration, and participation in scholarly and co-curricular activities.

PEO 3: Develop critical thinking, analytical reasoning, technical competence, and effective communication skills in which students can take responsibility as independent professionals or collaborative team members in academic, industrial, clinical, and research environments.

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PEO 4: Uphold ethical values, professional integrity, social responsibility, and environmental stewardship while contributing to sustainable development and critical outlook of diversified professional services based on social responsiveness and awareness as Biochemists.

Program Learning Outcomes (PLOs)

Upon successful completion of the B.Sc. (Honours) in Biochemistry and Molecular Biology program, graduates will be able to:

PLO 1: Knowledge

Demonstrate comprehensive knowledge and understanding of the fundamental principles of biochemistry, molecular biology, cell biology, genetics, microbiology, chemistry, and related life sciences.

PLO 2: Scientific Inquiry and Problem Solving

Apply biochemical and molecular biological principles, scientific reasoning, and critical thinking to identify, analyze, and solve biological, biomedical, environmental, and industrial problems.

PLO 3: Laboratory and Technical Skills

Perform biochemical and molecular biology laboratory techniques safely and accurately; operate standard laboratory instruments; and analyze, interpret, and report experimental data using appropriate scientific methods.

PLO 4: Research and Innovation

Design, conduct, and communicate scientific investigations by applying appropriate research methodologies, statistical analysis, bioinformatics tools, and ethical research practices.

PLO 5: Communication and Teamwork

Communicate scientific concepts, research findings, and technical information effectively through oral presentations, scientific writing, and digital media, while working efficiently as an individual and as a member of multidisciplinary teams.

PLO 6: Professionalism and Ethics

Demonstrate professionalism, integrity, ethical responsibility, biosafety, biosecurity, and respect for legal and ethical standards in education, research, and professional practice.

PLO 7: Leadership and Entrepreneurship

Apply leadership, management, and entrepreneurial skills to contribute to biotechnology, healthcare, pharmaceuticals, food industries, environmental management, and other life science sectors.

PLO 8: Lifelong Learning and Social Responsibility

Engage in lifelong learning, adapt to emerging scientific and technological developments, and contribute responsibly to society through sustainable, ethical, and evidence-based practices.

Mapping of PEOs with PLOs

PEOs/PLOs	PEO-1	PEO-2	PEO-3	PEO-4
PLO-1	✓	✓		
PLO-2	✓	✓	✓	
PLO-3	✓	✓	✓	
PLO-4	✓	✓	✓	✓
PLO-5		✓	✓	✓
PLO-6			✓	✓
PLO-7	✓	✓	✓	✓
PLO-8		✓	✓	✓

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Courses, Marks & Credit Distribution

SECOND YEAR

Course Code	Course Title	Marks	Credits
8222951	Enzymes & Bioenergetics	100	4
8222953	Carbohydrate Metabolism	100	4
8222955	Molecular Biology-I	100	4
8222958	Biochemistry Lab-II	100	4
8223057	Botany-II	100	4
8223058	Botany Lab-II	50	2
8223157	Zoology-II	100	4
8223158	Zoology Lab-II	50	2
8226601	Applied Information and Communication Technology (Applied ICT)	75	3
8226602	Practical: (Applied ICT)	25	1
8221171	Functional English	100	4
8222960	Viva-Voce (Biochemistry and Molecular Biology)	50	2
	Total	**950**	**38**

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Enzymes & Bioenergetics			
Course Code	**Marks**	**Credits**	**Contact Hours**
8222951	100	4	56

Course Objectives (COs): This course aims to provide students with a comprehensive understanding of enzyme structure, function, kinetics, regulation, and catalytic mechanisms, as well as the fundamental principles of bioenergetics and biological energy transformation. It enables students to analyze enzyme-catalyzed reactions, interpret enzyme behavior under different physiological conditions, and apply enzymological and bioenergetic concepts to metabolism, biotechnology, clinical diagnostics, and biomedical research.

Course Learning Outcomes (CLOs)

After successful completion of the course, students will be able to:

CLO	Course Learning Outcome	Bloom's Level
CLO1	Recall the EC classification and nomenclature of enzymes, enzyme activity units, kinetic parameters (Km, Vmax, kcat) and high-energy phosphate compounds.	Remember
CLO2	Explain the effects of substrate, enzyme, pH, temperature, cofactors and metal ions on enzyme activity, the catalytic mechanisms of chymotrypsin and ribonuclease A, and the MWC and Koshland models of allostery.	Understand
CLO3	Calculate specific activity, Km, Vmax and catalytic efficiency from Michaelis–Menten and Lineweaver–Burk plots, and $\Delta G^{\circ\prime}$ from equilibrium constants and redox potentials.	Apply
CLO4	Differentiate types of enzyme inhibition and ordered, random and ping-pong bisubstrate mechanisms from kinetic data.	Analyze
CLO5	Assess the clinical, pharmaceutical and biotechnological significance of enzyme inhibitors, catalytic antibodies (Isoenzymes), ribozymes and novel enzymes.	Evaluate

Mapping of CLOs with PLOs

Strength of alignment: 3 = Strong, 2 = Moderate, 1 = Low, blank = no alignment.

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8
CLO1	3							
CLO2	3	1						
CLO3	2	3		1				
CLO4		3		2				
CLO5		3						2

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Course Contents

Week	Course Content	Teaching–Learning Strategies	Assessment Strategy	Related CLO(s)
1-2	**Introduction to Enzymology:** History of enzymology; enzymes as biological catalysts; International enzyme classification (EC system); nomenclature; enzyme assay; enzyme activity; specific activity; enzyme activity units.	Interactive lectures, multimedia presentations, class discussion	Class participation, Quiz	CLO1, CLO3
3-4	**Factors Affecting Enzyme Activity:** Effect of substrate concentration, enzyme concentration, pH, temperature, cofactors, coenzymes, and metal ions on enzyme activity.	Interactive lectures, demonstrations, tutorials	Quiz, Assignment	CLO2
5-7	**Enzyme Kinetics:** Michaelis–Menten equation; Lineweaver–Burk plot; determination and significance of Km and Vmax; catalytic efficiency (overview).	Lectures, numerical problem-solving	Class test, Tutorial	CLO1, CLO3, CLO4
8-11	**Enzyme Inhibition:** Competitive, non-competitive, uncompetitive and irreversible inhibition; mechanism-based inhibition; pharmaceutical and clinical applications.	Lectures, case-based learning, problem-solving	Case analysis, Quiz	CLO4, CLO5
12-15	**Enzyme Structure and Catalytic Mechanisms**: Functional groups essential for catalysis; active site; enzyme–substrate complex; enzyme specificity; acid–base, covalent, proximity and orientation, and bond strain catalysis; chymotrypsin and ribonuclease A.	Interactive lectures, molecular models, tutorials	Mid-term examination, Assignment	CLO2
16-19	**Enzyme Regulation:** Allosteric enzymes; cooperativity; Monod–Wyman–Changeux (MWC) model; Koshland model; covalent modification; isoenzymes (ATCase, glycogen phosphorylase, lactate dehydrogenase).	Lectures, concept mapping, group discussion	Seminar presentation, Quiz	CLO2, CLO5
20-22	**Bisubstrate Reactions and Novel Enzymes:** Ordered and random sequential reactions; ping-pong reactions; catalytic antibodies (Isoenzymes); ribozymes; applications of novel enzymes	Lectures, seminar presentations, literature review	Presentation	CLO4, CLO5
23-28	**Bioenergetics:** Thermodynamics of biochemical reactions; Gibbs free energy; relationship between free energy and redox potential; ATP cycle; high-energy phosphate compounds; phosphate transfer reactions; ATP as the universal energy currency, Redox reaction, Biochemical couple reaction	Interactive lectures, problem-solving sessions	Class test, Final examination	CLO1, CLO3

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Course Assessment Framework

Total Course Marks: 100 Examination Duration: 4 Hours

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	In-course Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

Books Recommended

Berg, J. M., Tymoczko, J. L., Gatto, G. J., & Stryer, L. (2019). *Biochemistry* (9th ed.). W. H. Freeman.

Iwasa, J., & Marshall, W. (2020). *Karp's cell and molecular biology* (9th ed.). John Wiley & Sons.

Nelson, D. L., Cox, M. M., & Hoskins, A. A. (2021). *Lehninger principles of biochemistry* (8th ed.). W. H. Freeman; Macmillan Learning.

Voet, D., & Voet, J. G. (2016). *Biochemistry* (5th ed.). John Wiley & Sons.

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Carbohydrates Metabolism			
Course Code	**Marks**	**Credits**	**Contact Hours**
8222953	100	4	56

Course Objectives (COs): This course provides a comprehensive understanding of carbohydrate metabolism, emphasizing the biochemical pathways involved in energy production, storage, and regulation. Students will explore glycolysis, gluconeogenesis, glycogen metabolism, the pentose phosphate pathway, the citric acid cycle, and their hormonal regulation and electron- transport chain. The course also introduces metabolic integration and disorders associated with carbohydrate metabolism, enabling students to understand the biochemical basis of normal physiology and disease.

Course Learning Outcomes (CLOs)

After successful completion of the course, students will be able to:

CLO	Course Learning Outcome	Bloom's Level
CLO1	List the reactions, enzymes, irreversible (regulatory) steps and products of glycolysis, gluconeogenesis, glycogen metabolism, the pentose phosphate pathway and the TCA cycle, ETC	Remember
CLO2	The fundamental concepts of metabolism, including metabolic pathways, energy transfer, and the integration of carbohydrate metabolism. Explain allosteric and hormonal (insulin, glucagon, epinephrine) regulation of carbohydrate metabolism, the reciprocal regulation of glycolysis and gluconeogenesis, and the Cori and glucose–alanine cycles,ETC, Thermogenesis, Mitocondrial protein	Understand
CLO3	Calculate the net yield of ATP and reduced coenzymes from aerobic and anaerobic glycolysis and the TCA cycle.	Apply
CLO4	Analyze the integration of carbohydrate pathways, including the amphibolic and anaplerotic roles of the TCA cycle, in fed, fasting, starvation and exercise states. chemiosmosis properties	Analyze
CLO5	Evaluate clinical case data to justify the biochemical basis of diabetes mellitus, hypoglycemia, lactic acidosis, G6PD deficiency, galactosemia, hereditary fructose intolerance and glycogen storage diseases.	Evaluate

Mapping of CLOs with PLOs

Strength of alignment: 3 = Strong, 2 = Moderate, 1 = Low, blank = no alignment.

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8
CLO1	3							
CLO2	3	1						
CLO3	2	3						
CLO4	2	3						
CLO5		3						2

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Course Contents

Week	Course Content	Teaching–Learning Strategies	Assessment Strategy	Related CLO(s)
Week 1	**Introduction to Carbohydrate Metabolism**: Biological importance of carbohydrates, overview of metabolism: catabolism and anabolism, ATP as the energy currency, metabolic pathways, metabolic regulation.	Interactive lectures, multimedia presentations, class discussion	Class participation, Quiz	CLO1, CLO2
2-6	**Glycolysis:** Glycolytic pathway, energetics, irreversible steps, regulation of glycolysis, Fermentation, aerobic and anaerobic glycolysis, physiological significance	Interactive lectures, pathway analysis, problem-solving, case-based learning	Class test, Assignment, Tutorial	CLO1, CLO3, CLO5
7-10	**Gluconeogenesis**: Precursors, bypass reactions, regulation, Cori cycle, glucose–alanine cycle, reciprocal pathway and regulation with glycolysis.	Interactive Lectures, tutorials, pathway mapping, group discussion	Quiz, Problem-solving exercises	CLO1, CLO2
11-14	**Glycogen Metabolism**: Glycogenesis, glycogenolysis, hormonal regulation (insulin, glucagon, epinephrine), physiological significance.	Interactive lectures, case studies, group discussion	Mid-term examination, Case study	CLO2, CLO5
15-16	**Pentose Phosphate Pathway**: Oxidative and non-oxidative phases, NADPH production, ribose-5-phosphate synthesis, physiological importance, G6PD deficiency, glyoxalate cycle.	Interactive Lectures, clinical case discussion, tutorial	Quiz, Assignment	CLO1, CLO5
17-18	**Integration and Regulation of Carbohydrate Metabolism:** Metabolic integration in fed and fasting states, hormonal regulation, metabolic adaptation during starvation and exercise.	Lectures, concept mapping, problem-based learning	Group Discussion, Short Presentation	CLO3, CLO4
19-22	**Citric Acid (TCA) Cycle**: Reactions, regulation, energetics, amphibolic role, anaplerotic reactions, metabolic significance.	Interactive lectures, pathway integration, problem-solving	Class test, Tutorial	CLO2, CLO3, CLO4
23-26	**Electron transport Chain and oxidative phosphorylation** : the evolution of electron-transport chain, oxidation-reduction enzymes and electron transport, oxidative phosphorylation, uncouplers and inhibitors of oxidative phosphorylation, mitochondrial structure and the compartmentation of respiratory metabolism; disorders due to deficiencies of mitochondrial protein, brown adipose tissue and thermogenesis, mechanism of thermos-genesis. Role of addipose tissue	Interactive Lectures, concept mapping, problem-based learning	Group discussion, short presentation	CLO2, CLO4
27-28	**Clinical Aspects of Carbohydrate Metabolism:** Diabetes mellitus, hypoglycemia, lactic acidosis, galactosemia, hereditary fructose intolerance, lactose intolerance (overview), metabolic syndrome (introduction),glycogen storage diseases, mitochondrial protein related diseases.	Case-based learning, student presentations, discussion	Case analysis, Final examination	CLO5

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Course Assessment Framework

Total Course Marks: 100 Examination Duration: 4 Hours

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	In-course Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

Books Recommended

Berg, J. M., Tymoczko, J. L., Gatto, G. J., & Stryer, L. (2019). *Biochemistry* (9th ed.). W. H. Freeman.

Iwasa, J., & Marshall, W. (2020). *Karp's cell and molecular biology* (9th ed.). John Wiley & Sons.

Nelson, D. L., Cox, M. M., & Hoskins, A. A. (2021). *Lehninger principles of biochemistry* (8th ed.). W. H. Freeman; Macmillan Learning.

Voet, D., & Voet, J. G. (2016). *Biochemistry* (5th ed.). John Wiley & Sons.

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Molecular Biology-I			
Course Code	**Marks**	**Credits**	**Contact Hours**
8222955	100	4	56

Course Objectives (COs): This course aims to provide students with a comprehensive understanding of the molecular basis of heredity, the structure and properties of nucleic acids, and the molecular mechanisms of DNA replication, transcription, and translation. It develops knowledge of genome organization, gene regulation, recombinant DNA technology, DNA sequencing, and bioinformatics while enabling students to analyze molecular processes, interpret experimental data, and apply molecular biology principles to biological research, biotechnology, medicine, and clinical problem-solving.

Course Learning Outcomes (CLOs)

After successful completion of the course, students will be able to:

CLO	Course Learning Outcome	Bloom's Level
CLO1	Recall Mendel's laws, the landmark experiments establishing DNA as the genetic material, and the chemical components and conformations (A-, B-, Z-DNA) of nucleic acids.	Remember
CLO2	Explain the mechanisms of DNA replication, transcription and translation in prokaryotes and eukaryotes, and the organization of chromatin and chromosomes.	Understand
CLO3	Calculate Tm and base composition, deduce DNA sequences from Sanger sequencing data, and retrieve and annotate sequences from genomic databases.	Apply
CLO4	Interpret experimental evidence from landmark experiments, denaturation–renaturation and Cot curves, and autoradiography of DNA replication.	Analyze
CLO5	Justify the choice of cloning vectors (plasmid, cosmid, bacteriophage, BAC, YAC), library types and sequencing methods, and assess the biomedical significance of protein synthesis inhibitors.	Evaluate

Mapping of CLOs with PLOs

Strength of alignment: 3 = Strong, 2 = Moderate, 1 = Low, blank = no alignment.

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8
CLO1	3							
CLO2	3	1						
CLO3		3		2				
CLO4		3		2				
CLO5		3					1	

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Course Contents

Week	Course Content	Teaching–Learning Strategies	Assessment Strategy	Related CLO(s)
1-2	**Introduction to Molecular Biology:** Gene concept; relationship between genes, chromosomes, DNA as the genetic material; Application of molecular biology, Central dogma	Interactive lectures, multimedia presentations, group discussion	Class participation, Quiz, Short assignment	CLO1, CLO4
3-6	**Chemistry and Structure of Nucleic Acids:** Classification and composition of nucleic acids; nitrogenous bases, sugars, nucleosides, nucleotides; Watson–Crick model; physicochemical properties of DNA; DNA denaturation and renaturation; Tm value; Cot curve; hybridization kinetics; DNA conformations (A-, B-, and Z-DNA); repetitive DNA; palindromic sequences; cruciform DNA;	Interactive lectures, illustrations, tutorial sessions	Quiz, Tutorial, Class test	CLO1, CLO3, CLO4
7-11	**Genome Organization and DNA Packaging:** Genome organization; chromosome structure; nucleosome; chromatin organization; euchromatin and heterochromatin; interrupted genes;	Interactive lectures, concept mapping, discussion	Quiz, Group discussion	CLO2
12-16	**DNA Replication:** Semiconservative and semi discontinuous replication; bidireactional, unidireactional, and rolling-circle replication; enzymes involved (DNA polymerases, helicases, ligases, topoisomerases, primases, nucleases, methylases);	Interactive Lectures, animations, problem-solving sessions	Problem-solving exercises, Mid-term	CLO2, CLO4
17-20	**Transcription:** Genetic code; codon–anticodon interaction; transcription in prokaryotes and eukaryotes; RNA polymerases; promoters; enhancers; transcription factors; transcription activators; RNA processing (5′ capping, splicing, polyadenylation); mRNA transport;	Interactive lectures, tutorials, case discussion concept mapping, discussion,	Quiz, Group discussion	CLO2
21-24	**Protein Biosynthesis:** Ribosome structure; initiation, elongation, and termination of translation; polysomes; post-translational modifications (overview); inhibitors of protein synthesis; regulation of translation; biomedical significance (Antibiotic action).	Interactive Lectures, animations, case-based learning, student presentation	Student presentation, Final examination	CLO2, CLO5
25-28	**DNA Sequencing and Recombinant DNA Technology:** Sanger and Maxam–Gilbert DNA sequencing methods; plasmids, cosmids, bacteriophages, BACs, YACs; cDNA and genomic libraries; screening of recombinant clones.	Interactive Lectures, demonstrations, literature review, seminar	Seminar presentation, Assignment	CLO3 CLO5

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Course Assessment Framework

Total Course Marks: 100 Examination Duration: 4 Hours

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	In-course Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

Books Recommended

Alberts, B., Heald, R., Johnson, A., Morgan, D., Raff, M., Roberts, K., & Walter, P. (2022). *Molecular biology of the cell* (7th ed.). W. W. Norton & Company.

Berg, J. M., Tymoczko, J. L., Gatto, G. J., & Stryer, L. (2019). *Biochemistry* (9th ed.). W. H. Freeman.

Glick, B. R., & Patten, C. L. (2017). *Molecular biotechnology: Principles and applications of recombinant DNA* (5th ed.). ASM Press.

Iwasa, J., & Marshall, W. (2020). *Karp's cell and molecular biology* (9th ed.). John Wiley & Sons.

Krebs, J. E., Goldstein, E. S., & Kilpatrick, S. T. (2017). *Lewin's Genes XII*. Jones & Bartlett Learning.

Nelson, D. L., Cox, M. M., & Hoskins, A. A. (2021). *Lehninger principles of biochemistry* (8th ed.). W. H. Freeman / Macmillan Learning.

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Biochemistry Lab II			
Course Code	**Marks**	**Credits**	**Contact Hours**
8222958	100	4	112

Course Objectives (COs): This laboratory course provides hands-on training in fundamental biochemical and microbiological techniques. Students will develop practical skills in biochemical analysis, solution preparation, quantitative estimation of biomolecules, synthesis of biologically important compounds, microbial culture techniques, and interpretation of experimental data. The course emphasizes laboratory safety, scientific record keeping, data analysis, and the application of laboratory techniques in biomedical, clinical, food, and environmental sciences.

Course Learning Outcomes (CLOs)

After successful completion of the course, students will be able to:

CLO	Course Learning Outcome	Bloom's Level
CLO1	State good laboratory practice, biosafety and ethical rules for handling chemicals, laboratory equipment and microorganisms.	Remember
CLO2	Explain the chemical and microbiological principles of the saponification value, iodine value, Biuret, Nelson–Somogyi, Jaffe, Gram staining and Kirby–Bauer methods.	Understand
CLO3	Prepare standard solutions and buffers, perform quantitative biochemical estimations and syntheses, and execute aseptic microbiological culture techniques.	Apply
CLO4	Analyze titration curves (pKa), standard curves, percentage yield and bacterial growth curves, and record the results in a laboratory notebook.	Analyze
CLO5	Evaluate the validity of experimental results against reference values and antibiotic susceptibility standards, identifying sources of error.	Evaluate

Mapping of CLOs with PLOs

Strength of alignment: 3 = Strong, 2 = Moderate, 1 = Low, blank = no alignment.

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8
CLO1	1					3		
CLO2	3		1					
CLO3			3			2		
CLO4		1	3		2			
CLO5		3		2				1

Course Contents

Week	Title of Experiment	Teaching–Learning Strategies	Assessment Strategy	Related CLO(s)
1-2	Laboratory safety, biosafety and laboratory ethics; preparation of standard solutions and buffers; determination of pKa of acetic acid.	Demonstration, hands-on practice	Observation, Laboratory notebook	CLO1, CLO3, CLO4
3-4	Determination of the saponification value of fats and oils.	Laboratory experiment, discussion	Practical performance, Viva voce	CLO2, CLO3

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Week	Title of Experiment	Teaching–Learning Strategies	Assessment Strategy	Related CLO(s)
5-6	Determination of the iodine value of fats and oils.	Hands-on experiment	Laboratory report, Practical performance	CLO2, CLO3
7-8	Determination of lactose content in milk.	Laboratory experiment	Laboratory report, Viva voce	CLO3, CLO4
9-10	Determination of protein concentration by the Biuret method.	Demonstration and practical	Practical performance, Viva voce	CLO2, CLO4
11-12	Determination of glucose concentration by the Nelson–Somogyi method.	Practical experiment	Laboratory notebook, Data analysis	CLO3, CLO4
13-14	Determination of creatinine concentration by the alkaline picrate (Jaffe) method.	Practical experiment	Practical performance, Viva voce	CLO3, CLO5
15-16	Determination of serum urea and uric acid by standard biochemical methods.	Practical experiment	Laboratory report, Viva voce	CLO3, CLO5
17-18	Isolation and estimation of cholesterol from egg using the ferric chloride–acetic acid method	Practical experiment	Laboratory report, Data interpretation	CLO3, CLO4
19-20	Preparation of slant and streak culture	Demonstration, hands-on microbiology laboratory	Practical performance, Viva voce, Laboratory notebook	CLO3, CLO4
21-24	Bacterial culture maintenance, and determination of bacterial growth curve (*E. coli*)	Demonstration, hands-on microbiology laboratory	Practical performance, Viva voce, Laboratory notebook	CLO1, CLO3, CLO4
26-28	Morphological studies of microorganisms by Gram staining.	Laboratory experiment, data interpretation, group discussion	Practical examination, Laboratory report, Viva voce	CLO3, CLO5

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Assessment Scheme (OBE)

Assessment Tool	Weight (%)
Viva	10
Laboratory Note	10
Final Practical Examination	80
Total	**100**

Books Recommended

Cappuccino, J. G., & Welsh, C. T. (2019). *Microbiology: A laboratory manual* (12th ed.). Pearson.

Plummer, D. T. (1987). *An introduction to practical biochemistry* (3rd ed.). Tata McGraw-Hill.

Varley, H., Gowenlock, A. H., & Bell, M. (1980). *Practical clinical biochemistry* (Vol. 1, 5th ed.). William Heinemann Medical Books.

Wilson, K., & Walker, J. (Eds.). (2018). *Principles and techniques of biochemistry and molecular biology* (8th ed.). Cambridge University Press.

Course Code	8223057	Marks:100	Credits: 4	Class Hours: 56
Course Title	Botany - II			

Course Description

This course provides an integrated understanding of fundamental concepts in cell biology, genetics, plant breeding, biotechnology, ecology, conservation biology, plant physiology, and phytochemistry. It examines cellular organization and division, inheritance and genetic variation, plant improvement and evolution, biotechnological applications, ecosystem structure and function, biodiversity conservation, plant physiological processes, and major plant biomolecules. The course emphasizes the application of botanical knowledge to plant growth, crop improvement, environmental management, biodiversity conservation, and sustainable development.

Course Objectives (COs)

After successful completion of this course, students will be able to:

1. Develop a comprehensive understanding of cell biology, genetics, plant breeding, and evolutionary principles underlying plant growth, reproduction, inheritance, and variation.
2. Explain the fundamental principles and applications of biotechnology in agriculture, industry, environmental management, plant improvement, and sustainable resource utilization.
3. Develop an understanding of ecological processes, environmental issues, ecosystem functioning, and biodiversity conservation, with emphasis on sustainable management of plant resources.
4. Apply knowledge of plant physiology and phytochemistry to explain plant growth, metabolism, productivity, adaptation, and the importance of major plant biomolecules

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Course Learning Outcomes (CLOs)

CLO	Course Learning Outcome (CLO)	Bloom's Taxonomy Level
CLO1	Explain the ultrastructure of plant cells, chromosome organization, and the mechanisms and significance of mitosis and meiosis.	Understand
CLO2	Explain the fundamental principles of inheritance, linkage, crossing over, mutation, and their roles in plant genetics and crop improvement.	Understand
CLO3	Describe the principles and applications of plant breeding, artificial pollination, selection methods, and theories of evolution.	Understand
CLO4	Explain the basic concepts and applications of biotechnology, including microbial fermentation, dairy biotechnology, and biogas production.	Understand
CLO5	Analyze ecological principles, ecosystem structure and function, ecological adaptations, biogeochemical cycles, environmental pollution, and greenhouse effects.	Analyze
CLO6	Explain the importance of biodiversity conservation and evaluate in-situ and ex-situ conservation strategies for sustainable management of plant resources.	Evaluate
CLO7	Explain the physiological processes of plants, including water relations, mineral nutrition, photosynthesis, respiration, enzyme action, photoperiodism, vernalization, and plant hormones.	Understand
CLO8	Describe the classification, structure, biological functions, and economic importance of major plant carbohydrates and communicate scientific concepts effectively through assignments and examinations.	Apply

Mapping of CLOs with PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2			√	√								
CLO3					√	√	√					
CLO4								√	√	√		
CLO5										√	√	√
CLO6	√	√	√									
CLO7				√	√							
CLO8						√	√					

Course Content

Unit Title	Detailed Unit Contents	Teaching–Learning Strategies	Assessment	CLO
Unit 1. Cytology Weeks 1–4	Ultra-structure of a eukaryotic cell, differences between prokaryotic and eukaryotic cells, physical and chemical composition of cell wall and plasma membrane. **Chromosome:** Physical and chemical nature, chromosome organization, types of	Interactive lectures, multimedia presentations, classroom discussion, diagrams	Quiz, class test, assignment, final examination	CLO1

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	chromosomes and role in heredity. **Cell division:** Mitosis and meiosis, stages and significance in growth, reproduction, genetic variation and evolution. **Cell organelles:** Structure and functions of nucleus, mitochondria, chloroplast, endoplasmic reticulum, ribosome, Golgi apparatus, vacuole and other cellular components.			
Unit 2. Genetic **Weeks 5–8**	**Basic concepts of genetics:** Gene, DNA and RNA structure, DNA replication, gene expression, Mendel's laws; linkage and crossing over; mutation, point mutation, physical and chemical mutagens and significance.	Lectures, multimedia presentations	Quiz, assignment, midterm, final examination	CLO1, CLO2
Unit 3. Plant Breeding and Evolution **Weeks 9–11**	**Plant Breeding and Evolution:** Definition and significance of plant breeding. Method of artificial pollination in plants. Selection method. **Theories of evolution:** Darwin's theory and synthetic theory of evolution.	Lectures, case studies, group discussion	Class presentation, assignment, written examination	CLO2
Unit 4. Biotechnology **Weeks 12–15**	Definition, concept, history and scope of biotechnology; applications in agriculture, industry, medicine and environmental management. **Milk products:** Techniques of production of curd and butter; role of microorganisms in dairy fermentation .**Biogas technology:** Production methods, microbial role, composition of biogas and its uses as renewable energy. **Plant tissue culture and its applications:** Principles, techniques, micropropagation and importance in crop improvement and conservation. **Biofertilizers and biopesticides:** Types, production, applications and importance in sustainable agriculture.	Interactive lectures, demonstrations, case studies	Quiz, assignment, final examination	CLO2
Unit 5. Plant Ecology and Environmental Science **Weeks 16–20**	Definition, differences between autecology and synecology. **Ecological factors:** Climatic, edaphic and biotic. **Ecological adaptation:** Characteristics of hydrophytes, xerophytes and halophytes. **Ecosystem:** Definition; pond ecosystem; food chain; food web; ecological pyramids. **Biogeochemical cycles:** Nitrogen and carbon cycles. **Pollution:** Air and water pollution. **Greenhouse effects:**	Lectures, field examples, case studies, group discussion	Assignment, presentation, midterm, final examination	CLO3

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	Definition, sources of greenhouse gases; causes and effects of greenhouse gases; CO_2, CFC, CH_4 and N_2O; ozone layer depletion; carbon dioxide and the world climate; control of greenhouse effects.			
Unit 6. Conservation Biology	**Conservation:** Importance of conservation of plant species. Methods of ex-situ and in-situ conservation.	Interactive lecture, case studies	Assignment, written examination	CLO3
Unit 7. Plant Physiology and Phytochemistry Weeks 21–28	**Water relation:** Mechanisms of absorption of water and minerals. **Mineral nutrition. Photosynthesis:** General account and modern concept. **Respiration:** Aerobic and anaerobic. **Enzymes:** Definition and properties. Photoperiodism. Vernalization**:** Definition and mechanism. **Plant hormones:** A general account. **Carbohydrates:** Classification with examples; structures of important monosaccharides and oligosaccharides and isomerism; storage polysaccharides; starch, inulin, glycogen; structural polysaccharides; cellulose and economic importance of carbohydrates.	Lectures, tutorials, problem-solving, multimedia presentations	Quiz, class test, assignment, midterm, final examination	CLO4, CLO5

Course Assessment Framework:

Total Course Marks: 100 Duration of Exam: 4 hours

Assessment Category	Assessment Component	Weightage
Formative Assessment (20%)	Attendance & Class Participation	5%
	In-course Examination	10%
	Quiz / Assignment Presentation	5%
Summative Assessment (80%)	Final Examination • Students will have to answer 8 questions out of 12 • 10 marks per question ($8 \times 10 = 80$ marks) • Each question may contain maximum 3 sub-sections (e.g., a, b, c)	80%

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List of Books

Agrios, G. N. (1997). *Plant pathology* (4th ed.). Academic Press, London, UK.

Lewin, B. (2000). *Gene 2000*. Oxford University Press & Cell Press, Oxford, UK.

Devlin, M. R., & Witham, H. F. (1986). *Plant physiology* (4th ed.). CBS Publishers and Distributors, New Delhi, India.

Esau, K. (1953). *Plant anatomy*. John Wiley & Sons, Inc., New York, USA.

Kumar, H. D. (1995). *General ecology*. Vikas Publishing House, New Delhi, India.

Pelczar, M. J., Chan, E. C. S., & Krieg, N. R. (1993). *Microbiology: Concepts and applications.* McGraw-Hill Book Company, New York, USA.

Vashishta, P. C. (1993). *Botany for degree students: Pteridophyta*. S. Chand & Company Ltd., New Delhi, India.

আখতারুজ্জামান, ম. (২০০০). *বিবর্তনবাদ*, হাসান বুক হাউজ, ঢাকা।

কর্মকার, যদুলাল. (২০০০) *উদ্ভিদ শরীর বিজ্ঞান*, হাসান বুক হাউজ, ঢাকা।

খন্দকার এম. (১৯৯৪). *ডলমনোলজী*, ঢাকা বিশ্ববিদ্যালয় প্রকাশনা, ঢাকা।

বাসার, এম. এ., এম.এ. হাসান এবং ম. রফিকুল ইসলাম. (২০০৪). *উদ্ভিদ বিজ্ঞান*, হাসান বুক হাউজ, বাংলা বাজার, ঢাকা।

হাসান, এম. এ. (১৯৯৬). *উাংলাদেশের ভেষজ উদ্ভিদ*, আশরাফিয়া বই ঘর, বাংলা বাজার, ঢাকা।

হাসান, এম. এ. এবং এম. কে. আলম. (১৯৯৭). *উদ্ভিদ শ্রেণী বিন্যাস* তত্ত (৩য় সংস্করণ), হাসান বুক হাউস, ঢাকা।

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Course Code	8223058	Marks:50	Credits: 2	Class Hours: 42
Course Title	Botany -II Practical			Time: 6 hours

Course Description

Course Objectives (Cos)

After completion of this course, students will be able to:

1. Develop practical skills in the identification, classification, and study of representative algae, fungi, bryophytes, pteridophytes, gymnosperms, and angiosperms through dissection, mounting, observation, and illustration.
2. Explain the morphological and anatomical characteristics of different plant groups and relate their structural adaptations to their ecological habitats.
3. Perform basic physiological experiments to understand fundamental plant processes such as plasmolysis, osmosis, and photosynthetic oxygen evolution.
4. Observe and identify different stages of mitotic cell division using plant materials and understand their significance in growth and development.
5. Develop skills in scientific observation, laboratory techniques, record keeping, scientific drawing, and interpretation of botanical specimens.

Course Learning Outcomes (CLOs)

CLO	Course Learning Outcome	Bloom's Taxonomy Level
CLO1	Identify and classify representative genera of algae, fungi, bryophytes, pteridophytes, gymnosperms, and angiosperms through practical study.	Understand
CLO2	Perform dissection, mounting, microscopic observation, scientific drawing, and description of plant specimens.	Apply
CLO3	Conduct basic plant physiology experiments and interpret the results related to water relations and photosynthesis.	Analyze
CLO4	Identify different stages of mitosis and explain their importance in plant growth and development.	Understand
CLO5	Analyze ecological adaptations of hydrophytic and xerophytic plants through morphological and anatomical observations.	Analyze
CLO6	Maintain laboratory records, prepare scientific reports, and demonstrate botanical knowledge through viva-voce.	Apply

Mapping of CLOs with PLOs

CLOs	PLO 1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√	√								

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CLO3				√	√	√	√					
CLO4	√	√		√				√	√			
CLO5		√				√	√		√	√		
CLO6	√	√	√								√	
CLO7				√	√					√		
CLO8					√	√						

Course Content

Unit Title	Detailed Unit Contents	Teaching–Learning Strategies	Assessment	CLO(s)
Unit 1: Plant Diversity, Identification and Systematics	1. Detailed study including dissection (where necessary), mounting, drawing, description and identification with classification of the following genera:Algae: *Chlamydomonas*, *Oedogonium* and *Vaucheria*.Fungi: *Yeast* and *Ascobolus*.Bryophyte: *Riccia* and *Marchantia*.Pteridophyte: *Selaginella*.Gymnosperms: *Cycas*.Angiosperm: Poaceae and Fabaceae.	Demonstration of specimens, laboratory observation, microscopic study, dissection and temporary slide preparation, scientific drawing practice, guided identification exercises.	Practical examination, specimen identification, laboratory performance, scientific drawing, practical record assessment.	CLO1 CLO2
Unit 2: Identification of Plant Genera with Diagnostic Characters	2. Identification of the following genera with reasons:Algae: *Volvox*, *Polysiphonea* and *Fucas*.Fungi: *Agaricus*, *Puccinia* and *Penicillium*.Lichen: *Crustose*, *Foliose* and *Fructose*.Bryophyte: *Anthoceros*.Pteridophyte: *Psilotum*, *Azolla* and *Pteris*.Gymnosperms: Male and female cones of *Cycas*.Angiosperms: Scientific names of common plants.	Specimen display, field-based observation, herbarium study, comparative analysis, identification keys and discussion.	Spot identification test, viva-voce, practical examination, herbarium/specimen report evaluation.	CLO1 CLO2
Unit 3: Plant Physiology Experiments	Experimental study of basic physiological processes:**(a)** Demonstration of plasmolysis: principle, experimental procedure, observation and precautions. **(b)** Demonstration of	Laboratory demonstration, hands-on experimentation, observation recording, data interpretation and	Experimental performance, observation note, practical report, viva-voce.	CLO3

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	osmosis: experimental setup, observation and interpretation. **(c)** Evolution of oxygen during photosynthesis: experimental demonstration and factors affecting photosynthetic activity.	discussion of physiological processes.		
Unit 4: Cytology and Cell Division	Study of different stages of mitosis in onion (*Allium cepa*) root tip: -Preparation of root tip squash. – Identification of stages: prophase, metaphase, anaphase and telophase. - Significance of mitosis in plant growth and development.	Microscopic observation, slide preparation demonstration, image-based learning, identification and interpretation of mitotic stages.	Microscopic identification , slide preparation assessment, practical examination and viva.	CLO4
Unit 5: Plant Ecology and Adaptation Studies	**(a)** Morphological and anatomical studies of hydrophytic and xerophytic plants. **(b)** Ecological identification of plants: – Identification of plants according to habitat. – Scientific naming and ecological significance.	Field observation, specimen collection, ecological comparison, anatomical study, group discussion and field-based learning.	Field report, plant identification test, practical record evaluation, viva-voce.	CLO5
Unit 6: Laboratory Documentation and Scientific Communication	Preparation and maintenance of laboratory notebook including: – Recording observations. – Scientific drawings and labeling. – Classification and description of specimens. – Preparation of practical reports.	Guided note preparation, feedback on scientific documentation, demonstration of proper laboratory practices.	Laboratory notebook evaluation, report assessment, continuous assessment.	CLO6
Unit 7: Viva-Voce and Practical Competency Evaluation	**Viva-Voce:** Assessment of students' understanding of: – Plant identification and classification. – Laboratory techniques. – Experimental principles and observations. – Ecological and cytological concepts.	Interactive questioning, specimen-based discussion, problem-solving and conceptual assessment.	Final practical examination, viva-voce, overall laboratory performance assessment.	CLO6

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List of Books

Agrios, G.N. (1997). *Plant Pathology* (4th ed.). Academic Press, London.
Benjamin L. (2000). *Gene 2000*. Oxford University Press and Cell Press. Bold, H.C. and
Kumar, H.D. (1995). *General Ecology*, Vikash Pub. House, India.
Lawrence, G.H.M. (1951). *Taxonomy of Vascular Plants*. The Macmillan Co. New York.

Pelczer, M. J., Chan, E. C. S. & Krieg, N. R. (1993). *Microbiology: Concepts and Applications*. McGraw Hill Book Co. Inc. New York.

Vashista, P.C. 1993). *Botany for Degree Students: Pteridophyta.* S.C. Chand & Co. Ltd. Ramnagar, New Delhi.

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কর্মকার, যদুলাল. (২০০০). *উদ্ভিদ শরীর বিজ্ঞান*, হাসান বুক হাউজ, ঢাকা।

খন্দকার মনিরুজ্জামান, (১৯৯৪). *লিমনোলজী*, ঢাকা বিশ্ববিদ্যালয় প্রকাশনা, ঢাকা।

বাসার, এম. এ., এম.এ. হাসান এবং ম. রফিকুল ইসলাম. (২০০৪). *উদ্ভিদ বিজ্ঞান*, হাসান বুক হাউজ, বাংলা বাজার, ঢাকা।

হাসান, এম. এ. (১৯৯৬). *বাংলাদেশের ভেষজ উদ্ভিদ*, আশরাফিয়া বই ঘর, বাংলা বাজার, ঢাকা।

হাসান, এম. এ. এবং এম. কে. আলম. (১৯৯৭). *উদ্ভিদ শ্রেণী বিন্যাস তত্ত* (৩য় সংস্করণ), হাসান বুক হাউস, ঢাকা।

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Instruction to the Examiners

Time: 6 hours Marks: 50

1. Detailed Study, Dissection, Mounting, Description and Identification of Plant Genera
Marks: 12 (Slide preparation/mounting: 2+ Neat labeled diagrams: 2+ Identification: 0.5+ Classification and identifying characters: 1.5=6×2=12)
Specimen A and B:
Two specimens will be provided from the following groups:

- Algae: *Chlamydomonas, Oedogonium, Vaucheria*
- Fungi: Yeast, *Ascobolus*
- Bryophyta: *Riccia, Marchantia*
- Pteridophyta: *Selaginella*
- Gymnosperms: *Cycas*
- Angiosperms: Families Poaceae and Fabaceae

Students will perform:

- Dissection (where necessary)
- Preparation of slides/mounting
- Observation and identification
- Classification
- Description of diagnostic features
- Scientific drawings with proper labeling

2. Spot Identification of Plant Genera with Reasons Marks: 9
(Identification: 0.5+ Identifying Characteristics: 1.0=1.5×6=9)
Specimens C–H will be provided for identification from the following groups:
Algae: *Volvox, Polysiphonia, Fucus*; Fungi: *Agaricus, Puccinia, Penicillium*
Lichens: Crustose, Foliose, Fruticose; Bryophyta: *Anthoceros*
Pteridophyta: *Psilotum, Azolla, Pteris*
Gymnosperms: Male and female cones of *Cycas*;
Angiosperms: Common flowering plants with scientific names
3. Plant Physiology Experiments Marks: 6
(Experimental setup and procedure: 3+ Observation and interpretation: 2 + Precautions and conclusion: 1= 6)
Students will perform experiments related to:
(a) Demonstration of Plasmolysis: Principle, Experimental procedure, Observation, Precautions
(b) Demonstration of Osmosis: Experimental setup, Observation and interpretation
(c) Evolution of Oxygen During Photosynthesis: Experimental demonstration, Observation of oxygen release, Factors affecting photosynthetic activity

4. Cytology: Study of Mitosis in Onion Root Tip Marks: 6
(Preparation of slide: 2+ Identification of mitotic stages: 2+ Neat labelled diagrams: 1+ Significance/comment:1=06)
Students will prepare and study onion (*Allium cepa*) root tip squash.
Stages to be studied: Prophase, Metaphase, Anaphase, Telophase

5. Plant Ecology and Adaptation Study Marks: 4

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Specimen I and J (Identification:0.5+ Identifying characters/adaptive features:1.5=2×2=4)
Specimens will be selected from:

- Hydrophytic plants
- Xerophytic plants

Students will study:

- Morphological adaptations, Anatomical adaptations
- Ecological significance, Identification of plants with scientific names

6. Collection, Laboratory Record and Viva-Voce Marks: 13

Components	Marks
Collection of plant specimens (Algae, Fungi, Bryophyta, Pteridophyta, Gymnosperms and Angiosperms)	3.0
Practical Laboratory Notebook	5.0
Viva-Voce	5.0
Total	13.0

Total Marks Distribution

Examination Component	Marks
Detailed study of specimens (A & B)	12
Spot identification (C–H)	9
Plant physiology experiment	6
Cytology (Mitosis)	6
Ecology and adaptation study	4
Collection	3
Practical Notebook	5
Viva-Voce	5
Grand Total	50

General Instructions

- Students must submit properly prepared practical notebooks containing classification, descriptions, observations and labelled drawings.
- Slides must be prepared carefully following standard laboratory techniques.
- Identification answers must include appropriate diagnostic characters.
- Scientific names should be written according to accepted botanical nomenclature.
- Herbarium specimens and field collections must include proper labels.
- Viva-voce will evaluate students' understanding of practical techniques, specimen identification and botanical concepts.

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Course Code	8223157	Marks: 100	Credits: 4	Class Hours: 56
Course Title	Zoology-II			Exam Duration: 4 Hours

Course Description

This course provides an integrated understanding of chordate diversity, comparative anatomy, evolution, palaeontology, zoogeography, ecology, genetics, animal breeding, and applied zoology. It emphasizes taxonomic classification, structural and functional adaptations, biodiversity and conservation, genetic principles, breeding techniques, and the applications of zoological knowledge in fisheries, poultry, agriculture, and sustainable resource management.

Course Objectives

1. To develop a comprehensive understanding of chordate diversity, classification, comparative anatomy, evolutionary relationships, and palaeontological evidence.
2. To develop knowledge of zoogeography, marine ecology, biodiversity, natural resources, conservation, and animal adaptations to different environments.
3. To develop an understanding of genetic principles, inheritance, mutation, population genetics, animal breeding methods, and modern breeding techniques.
4. To apply zoological knowledge to agricultural pest management, fisheries, integrated farming, poultry production, and sustainable resource utilization in the socio-economic context of Bangladesh.

Course Learning Outcome (CLOs)

CLO	Course Learning Outcome	Bloom's Taxonomy Level
CLO1	Classify chordate taxa (Chondrichthyes to Mammalia) up to the order level by applying diagnostic morphological characteristics, and evaluate the evolutionary significance of transitional groups through palaeontological evidence.	Analyze / Evaluate
CLO2	Compare and contrast the organ systems (skeletal, digestive, circulatory, respiratory, excretory, nervous, reproductive, and endocrine) across 9 type study organisms (Ascidia* to Cavia) to explain anatomical adaptations to diverse habitats and modes of life.	Analyze
CLO3	Analyze zoogeographical regions of the world, marine ecological zonations, and biodiversity concepts to assess conservation challenges and sustainable management of natural resources.	Analyze / Evaluate
CLO4	Apply principles of Mendelian genetics, multiple alleles, mutations, and chromosomal sex determination to solve inheritance problems and explain animal breeding strategies.	Apply
CLO5	Evaluate the principles and practices of applied zoology including agricultural pest management, integrated fish farming, and poultry farming and propose solutions for enhancing productivity and socio-economic conditions in	Evaluate / Create

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Mapping of CLOs with PLOs

CLO / PLO	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO 1	✓	✓							✓			
CLO 2	✓	✓										
CLO 3	✓	✓	✓	✓						✓		
CLO 4	✓	✓	✓								✓	
CLO 5	✓	✓	✓							✓	✓	

Course Content

Topic	Teaching Learning Strategy	Assessment	CLOs
<u>Group – A:</u> Chordates: (50 Marks) 1. Broad classification of the following up to order with general and diagnostic characteristics of each taxonomic category with examples (preferably local) and affinities- Chondriththyes, Osteichthyes, Amphibia, Reptilia, Aves and Mammalia. 2. Type study of the following with their systematic position, habitats, distribution, external morphology, anatomy including skeletal, digestive, circulatory, respiratory, excretory, nervous, reproductive and endocrine systems; food & feeding habits and mode of life - *a.* Urachordata : *Ascidia* *b.* Cephalochordata : *Branchiostoma* *c.* Cephalaspimorpha: *Petromyzon* *d.* Chondrichthyes : *Scoliodon* *e.* Osteichthyes : *Labeo rohita* *f.* Amphibia : *Bufo* *g.* Reptilia : *Hemidactylus* *h.* Aves : *Columba livia* *i.* Mammalia : *Cavia porcellus.*	Lectures and Group discussions Fossil Timeline Jigsaw Virtual Dissection Lab Organ System Expert Jigsaw	Class test, specimen-based identification, comparative anatomy assignment, virtual dissection activity, group presentation, written examination	CLO1 CLO2
<u>Group – B</u> (50 Marks) **Palaeontology, Zoogeography, Ecology, Genetics & Animal Breeding and Applied Zoology:** **Palaeontology:** Geologocal time table; extinction events of Mesozoic reptiles and dinosaurs; palaeontological history of Horse & Man; microfossil types and their structures.	Lecture, Group Discussion, Assignments and tutorial sessions Zoogeographical Mapping	Quiz, map-based assessment, case-study analysis, problem-solving	CLO3 CLO4 CLO5

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Zoogeography: Zoogeographical regions and sub-regions of the world, their boundaries, physical characteristics, climatic conditions, flora and fauna. **Ecology:** Marine ecology; Zonation and Animal Adaptation to Different Zonations; Concept of Biodiversity; Concept and Classification of Resources; Conservation of Natural Resources; Renewable and Non-Renewable Resources. **Animal Genetics:** Laws of inheritance, gene interactions and frequency, genotype frequency, Hardy-Weinberg Law, Mutation, genetic drift, Continuous variation, heritability, and repeatability of traits. **Animal Breeding:** Pedigree, progeny testing; Inbreeding, outbreeding, crossbreeding, and grading up; Selection response, selection index, and breeding value estimation (EBV); Marker-assisted selection and genomic selection in livestock. **Applied Zoology:** 1. **Major Branches:** Entomology, Fisheries, Wildlife Biology, Parasitology and others. 2. **Agricultural Pests:** Major Pests of Rice, Jute, Sugarcane & stored grain. 3. **Integrated Fish Farming:** Types; Poultry, Livestock and Paddy-cum -Fish Culture. 4. **Poultry farming:** System of poultry farming, diseases management, economic impacts.	Marine Zonation Poster Punnett Square Problem-Solving Mutation Case Studies Integrated Farming System Design Pest Identification & Management Plans	exercises, assignment, group presentation, farming-system design, pest-management plan, class test and written examination	

List of Books

Hickman, C. P., Jr., Keen, S. L., Eisenhour, D. J., Larson, A., & I'Anson, H. (2024). *Integrated principles of zoology* (19th ed., International student ed.). McGraw Hill.

Jordan, E. L., & Verma, P. S. (2022). *Chordate zoology* (14th ed.). S. Chand and Company.

Kardong, K. V. (2019). *Vertebrates: Comparative anatomy, function, evolution* (8th ed.). McGraw-Hill Education.

Klug, W. S., Cummings, M. R., Spencer, C. A., Palladino, M. A., & Killian, D. J. (2021). *Concepts of genetics* (12th ed.). Pearson.

Pough, F. H., Janis, C. M., & Heiser, J. B. (2013). *Vertebrate life* (9th ed.). Pearson.

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Begon, M., & Townsend, C. R. (2021). *Ecology: From individuals to ecosystems* (5th ed.). Wiley-Blackwell.

Benton, M. J. (2014). *Vertebrate palaeontology* (4th ed.). Wiley-Blackwell.

Gupta, S. K. (2019). *Applied zoology*. Narendra Publishing House.

Hafez, E. S. E., & Hafez, B. (2013). *Reproduction in farm animals* (7th ed.). Wiley-Blackwell.

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Course Code	8223158	Marks: 50	Credits: 2	Class Hour 42
Course Title	Zoology Practical-II			

Course Description

This practical course develops hands-on skills in the identification, classification, comparative study, dissection, and microscopic examination of chordates. It includes museum specimens, permanent slides, comparative osteology, chordate dissection, temporary mounting, and freshwater microfauna studies, with emphasis on accurate observation, scientific drawing, laboratory techniques, and practical documentation.

Course Objectives

1. To develop practical skills in the identification, classification, and comparative study of representative chordate specimens, permanent slides, and vertebrate bones.
2. To develop competency in chordate dissection, temporary mounting, staining, microscopic observation, scientific drawing, and labeling techniques.
3. To develop practical skills in identifying freshwater plankton and benthic microfauna and interpreting their diagnostic characteristics through systematic observation and laboratory records.

Course Content

1. Study of Museum Specimens: Representative of Chordate phyla (at least 35 specimens to be studied)**.**

Chordates*:* *Ascidia mentula, Scoliodon laticaudus, Labeo rohita, Labeo calbasu, Catla catla, Cirrhinus mrigala, Channa striata, Channa punctata, Channa marulius, Clarias batrachus, Heteropneustes fossilis, Anabas testudineus, Wallago attu, Tenualosa Ilisha, Puntius sophore, Mystus vittatus, Pangasius pangasius, Ompok pabda, Mystus aoer, Mastacembelus armatus, Chanda nama, Monopterus cuchia, Labeo bata, Bufo melanostictus, Microhyla ornata, Rana tigrina, Naja kaouthia, Naja naja, Ophiophagus hannah, Python molurus, Calotes versicolor, Copsychus saularis, Corvus splendens, Acridotheres tristis, Passer domesticus, Halcyon smyrnensis, Columba livia, Macaca mulatta, Axis axis, Felis chaus, Platanista gangetica.*

2. Study of Permanent Slides: Whole mount, body parts, various cells & tissues and different larvae (at least 20 slides to be studied).

3. Study of Bones: Comparative study of the skeleton of amphibian, reptile, bird and mammal.

4. Dissection: Dissection of the following Chordate Specimens-

a. Bufo/ any Carp - Brain and Cranial Nerves.
b. Lata fish-Afferent and Efferent Blood Vessels
c. Lizard - Circulatory System.

5. Temporary mounting of scales and fresh water organisms (staining 2+ mounting 2+ drawing 2 + labeling 2)

6. Fresh Water Studies: Identification of plankton and benthic microfauna in fresh water samples.

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Distribution of Marks for Second Year Final Practical Examination

No.	Particulars			Marks
1.	**Dissection:** One chordate item (Dissection – 5, display – 2, drawing & labeling - 3)			10
2.	Temporary mounting (staining 2+ mounting 2+ drawing 2 + labeling 2)			08
3.	Spotting of museum specimens			
	a.	Vertebrate specimens	4 items × 2 = 08 marks	
	b.	Permanent slides	3 items × 2 = 06 marks.	
	c.	Bones	2 items × 2 = 04 marks.	
			9 items × 2.0	18
4.	Freshwater studies (2 species of micro fauna to be shown – identification and, characteristics 1) – 2 items × 2 = 4			04
5.	Practical Class Note Books =			06
6.	Oral Test			04
			Total	50

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Course Code	Course Title	Marks	Credits
8222960	Viva-Voce (Biochemistry and Molecular Biology)	50	2

Course Description

The Viva-Voce course is designed to comprehensively assess students' conceptual understanding, theoretical integration, laboratory competencies, and communication skills acquired throughout their second year of study. The examination evaluates student mastery across all major second-year departmental courses (**Enzymes & Bioenergetics, Carbohydrate Metabolism, Molecular Biology-I, and Biochemistry Lab-II**).

Furthermore, it assesses the student's ability to interface biological sciences with core allied subjects (**Botany-II and Zoology-II**), digital literacies (**Applied ICT**), and professional language skills (**Functional English**). The course tests critical thinking, analytical reasoning, scientific ethics, biosafety consciousness, and overall professional readiness for advanced life science research or industrial roles.

Course Objectives (COs)

1. **Synthesize Core Knowledge:** Assess the depth of students' theoretical understanding of second-year core subjects in enzymes, metabolism, molecular biology, and allied natural sciences.
2. **Evaluate Practical Competency:** Measure students' understanding of laboratory instrumentation, safety protocols, experimental logic, and troubleshooting in biochemistry and microbiology.
3. **Foster Scientific Communication:** Evaluate professional presentation, logical delivery, and scientific articulation using proper biochemical and medical terminology.
4. **Appraise Ethics and Safety:** Assess the student's mastery of biosafety, biosecurity, research integrity, and ethical standardizations within life sciences.
5. **Connect Global and National Contexts:** Examine students' ability to link biotechnology and molecular discoveries to healthcare advancements, clinical disease diagnosis, and national agricultural or environmental sustainability goals.

Course Learning Outcomes (CLOs)

Upon successful completion of this course, students will be able to:

- **CLO1:** Demonstrate a precise command of biochemical concepts, thermodynamics, metabolic sequences, and molecular structures using correct professional terminology.
- **CLO2:** Deduce kinetic parameters, explain regulatory mechanisms (hormonal/allosteric), and analyze experimental standard curves or biological replication mechanisms.
- **CLO3:** Correlate disruptions in metabolic or genetic processes with clinical case datasets (e.g., inborn errors of metabolism, diabetes, or antibiotic mutations).
- **CLO4:** Articulate practical solutions regarding laboratory standard solutions, aseptic techniques, safety barriers, and accurate diagnostic observations.
- **CLO5:** Integrate data from cross-disciplinary subjects (Botany, Zoology, ICT) to address broader issues in life sciences, biotechnology, and resource management.
- **CLO6:** Exhibit professional confidence, ethical reasoning, active listening, and concise verbal communication during formal academic evaluations.

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Matrix Framework: Core Focus Areas for Evaluation

To ensure an exhaustive assessment, the viva panel will examine students on the following course groups:

Group A: Core Departmental Courses (60% Weight)

- **Enzymes & Bioenergetics (8222951):** Enzyme classifications, kinetics (K_{m}), (V_{max}), Lineweaver-Burk plots), regulatory enzymes, mechanisms of catalysis (chymotrypsin/RNase A), and thermodynamic variables (Δ G, redox potentials).
- **Carbohydrate Metabolism (8222953):** Detailed pathways (Glycolysis, TCA cycle, Pentose Phosphate Pathway, Glycogen storage/lysis), ATP calculation yields, hormonal regulation loops, and critical conditions (Diabetes Mellitus, G6PD deficiency, Glycogen Storage Diseases).
- **Molecular Biology-I (8222955):** Structure and conformations of nucleic acids (A-, B-, Z-DNA), replication steps, transcription/RNA editing, prokaryotic/eukaryotic translation mechanisms, cloning vectors, and DNA sequencing methods (Sanger).
- **Biochemistry Lab-II (8222958):** Quantitative estimation mechanics (Biuret, Nelson-Somogyi, Jaffe methods), buffer dynamics, isolation of lipids, microbially aseptic streaks, Gram staining, and bacterial growth graphs.

Group B: Allied Natural Sciences (25% Weight)

- **Botany-II & Lab (8223057/8223058):** Plant cytology, tissue culture processes, biofertilizers, photosynthesis dynamics, economic importance of plant carbohydrates, and ex-situ/in-situ conservation parameters.
- **Zoology-II & Lab (8223157/8223158):** Comparative chordate anatomy, genetics problem solving (Mendelian/Hardy-Weinberg), applied entomology, and integrated aquaculture frameworks.

Group C: Language, Applied Tools & Technology (15% Weight)

- **Applied ICT & Practical (8226601/8226602):** Principles of biological databases, basic digital logic toolsets, and introductory data reporting application parameters.
- **Functional English (8221171):** Syntactical proficiency, conversational fluency, and the capacity to decipher and present high-level academic content fluently.

Teaching & Learning Strategies

The preparation for the viva-voce examination utilizes interactive, peer-driven, and faculty-guided exercises:

- **Interactive Orientation Sessions:** Faculty review maps detailing expectations, structure, and pacing strategies.
- **Core Topic Synopses & Concept Mapping:** Guided rapid-fire question-and-answer hours focusing on pathway integrations.
- **Simulated/Mock Viva Examinations:** Realistic practice scenarios run by departmental sub-committees followed by descriptive critique sessions.
- **Self-Directed Study Blocks:** Assigned reviews of laboratory documentation notes and experimental calculations.

Assessment Strategy & Evaluation Metrics

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The course concludes with a final oral board examination conducted directly by a designated Examination Committee. The **50 total marks** are distributed across four distinct professional performance categories:

Evaluation Parameter	Key Focus Points	Max Marks
1. Theoretical Depth & Knowledge Base	Accuracy in recalling classifications, calculations, clinical integrations, structural logic, and pathways.	**20**
2. Technical & Methodological Reasoning	Clarity regarding laboratory setups, biosafety logic, troubleshooting errors, and data calculations.	**15**
3. Analytical Fluency & Problem Solving	Capabilities in analyzing case files, reasoning under stress, and applying theory to cross-disciplinary scenarios.	**10**
4. Communication, Demeanor & Presentation	Use of correct professional terminology, responsiveness, vocal clarity, confidence, and adherence to time guidelines.	**05**
Grand Total		**50 Marks**

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